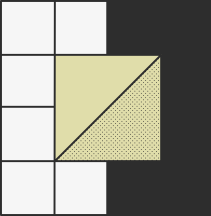


UNDERWATER GARBAGE DETECTION USING COMPUTER VISION

DAY-09

Kavya Hanishka





Dataset Findings

Dataset Audit:

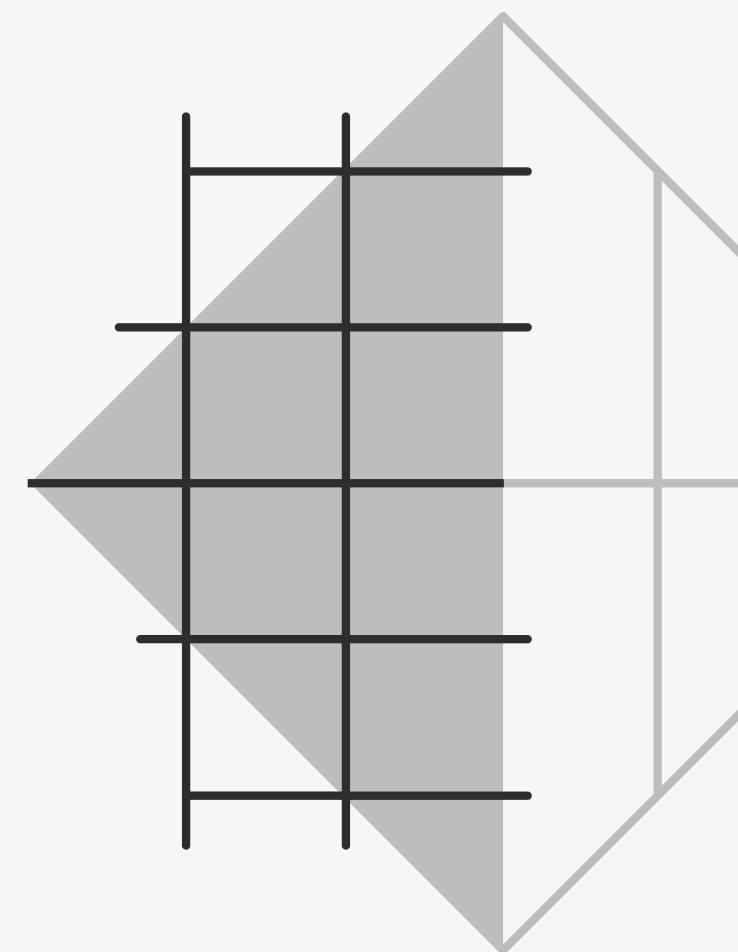
The dataset contains 7,214 JPG images and 7,212 annotation JSON files. After matching the filenames correctly, 7,212 image–annotation pairs were found. The annotations are in Supervisely JSON format and use bitmap geometry for the annotated objects.

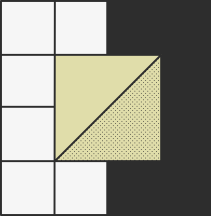
Quality Findings:

The initial integrity check found 0 corrupt images and 0 annotations without corresponding images. However, 2 images do not have annotations: vid_000270_frame0000041.jpg and vid_000270_frame0000044.jpg. The dataset also contains two resolutions: 480×270 (3,967 images) and 480×360 (3,247 images).

Key Finding:

Overall, the dataset is mostly structurally consistent. The two missing annotations need to be reviewed, and the different image resolutions need to be handled carefully during preprocessing.





Current Data Issues

Missing Annotations:

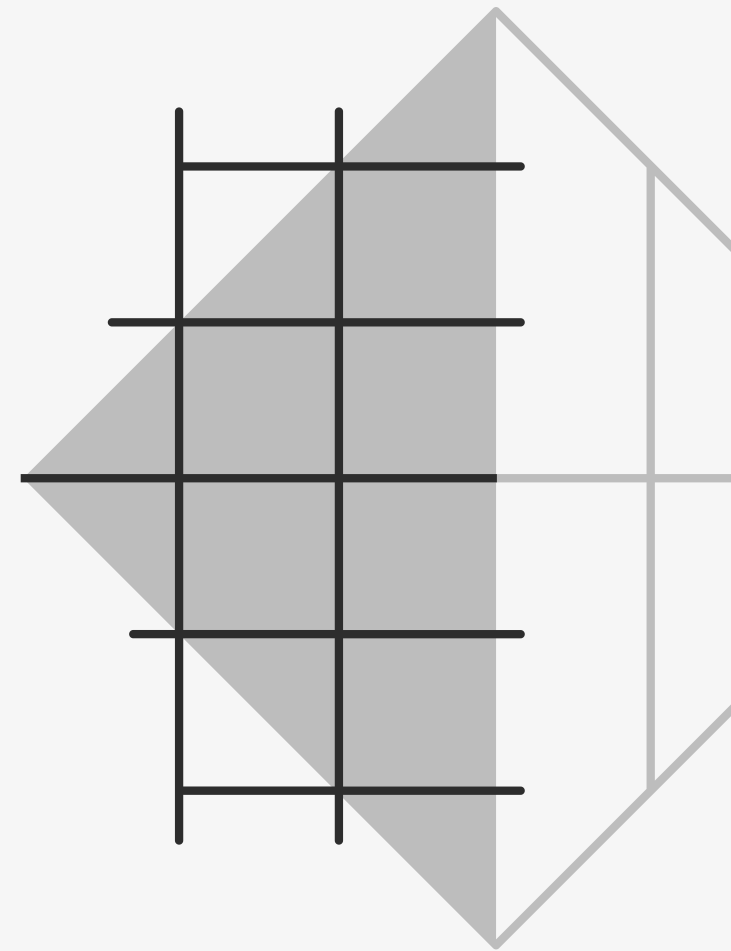
Two frames from vid_000270 do not have corresponding annotation files. These frames will be inspected manually before deciding whether the annotations should be recovered/created or whether the images should be excluded with a documented reason.

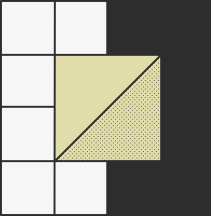
Different Resolutions:

The dataset contains 480×270 and 480×360 images. This is not necessarily an error, but directly resizing both to the same dimensions could distort objects. Therefore, an aspect-ratio-preserving resizing or letterboxing approach is proposed.

Annotation Format:

The original annotations use Supervisely JSON with bitmap geometry. The original annotation files will be preserved, and conversion to the final model format will be done only after confirming the modelling approach.





Proposed Split Strategy & Next Steps

Split Strategy:

A 70% train, 15% validation and 15% test split is proposed, with source/video-level splitting to reduce leakage from similar consecutive frames.

Preprocessing:

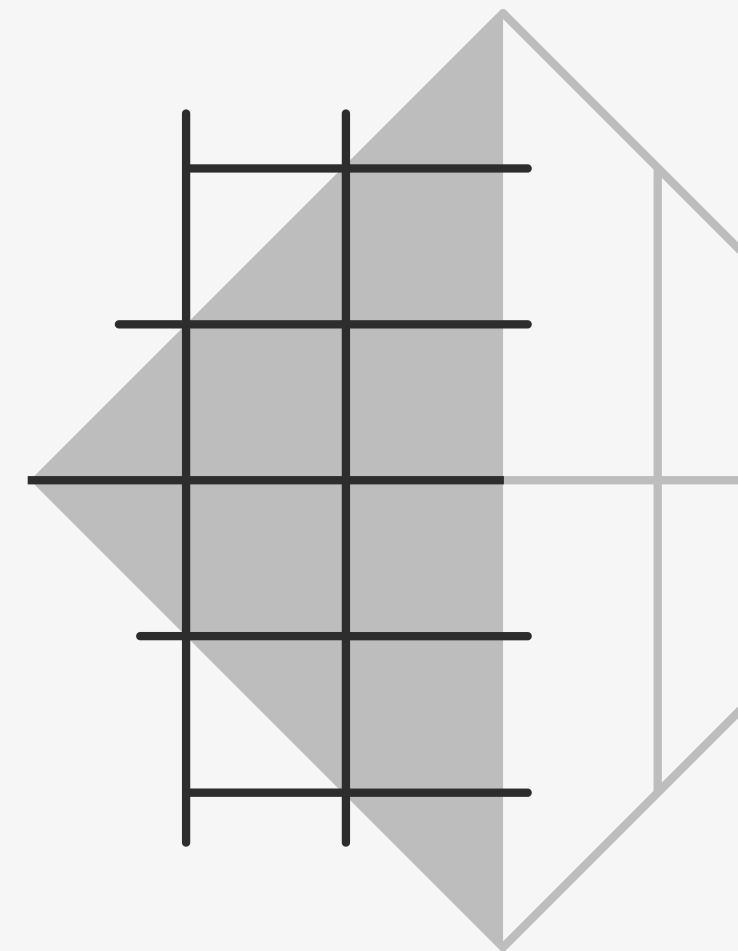
Use aspect-ratio-preserving resizing/letterboxing. Missing annotations will be reviewed and corrected where necessary.

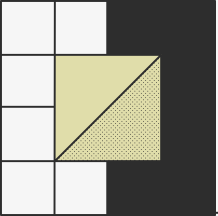
Annotation & Augmentation:

Preserve the original Supervisely annotations and convert them to the required YOLO format after confirming the final model. Augmentation will be applied only where useful.

Next Steps:

Visual EDA → Preprocessing → YOLO Baseline → Evaluation → Error Analysis





Thank you

